



Web programming and databases

Lab Exercise Sheet

Week 03 (PostgreSQL)

Instructions:

- All students are divided into two Teams A and B. Attend the relevant lab session for your attendance

Installation instructions

Lab machines (IT Cluster):

Open [AppsAnywhere](https://dubaiapps.bham.ac.uk) portal in the browser <https://dubaiapps.bham.ac.uk> and install “PostgreSQL+pgAdmin 4” by clicking on the “Launch” button.

Personal Laptops (Windows/Mac): Check Canvas [link](#) for installation instruction. Don't forget the Admin password. **Install version 14.7 or older as some commands will not run on latest version 15.2.**

Database server configuration

1) Access local database server

Open the program “psql” and/or “pgAdmin 4”

```
Server [localhost]:  
Database [postgres]:  
Port [5432]:  
Username [postgres]:  
Password for user postgres:
```

default password for local postgres instance on Dubai lab cluster machine is “postgresql”.

2) Access remote database server (Edgbaston)

Step 1: Check if your CS account exists, by visiting this link:

<https://my-account.cs.bham.ac.uk/>

and enter your University (ADF) username and password to login

Step 2:

Open the program “psql” and/or “pgAdmin 4”

Remote hostname: postgres-teaching.services.aws.cs.bham.ac.uk

Username: your University (ADF) username

Password: your CS account password (from Step 1)



Exercise 1: How can normalization be applied to eliminate redundancy in the provided table?

- First normal form (1NF)
- Second normal form (2NF)
- Third normal form (3NF)

Student_Grade_Report (StudentNo, StudentName, Major, CourseNo, CourseName, InstructorNo, InstructorName, InstructorLocation, Grade)

[Use either local host or remote hostname to do lab exercises]

Exercise 2. Create a postgresql database to store DVD rental information

In this exercise you are going to create a database to store all the required information for a DVD rental shop. This exercise is quite open, to allow you to explore not only using SQL but a little bit of database design as well. A model answer will be provided at the end of the week but it is likely that your answer might differ:

- Create a new database called "ye_old_video_shop"
- Create a new user that is the owner of this database.
- Create at least 5 tables to store various pieces of data that will be useful for the owner of the shop.
- You need to decide what tables (entities) make the most sense for this shop. You are allowed to create more than 5 if required.
- Create appropriate columns for each table.
- Make sure that each column has an appropriate data type. Try and use other data types besides those discussed in class.
- Make sure that each table has an appropriate primary key.
- Create a few SELECT statements to test whether you are storing useful data.
- Save all the CREATE, INSERT and SELECT scripts in a single text file so that you can run the SQL code again if needed.